

Shakiba Ahmadimehr

Biomedical Engineer & Data Analyst

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RESEARCH INTERESTS

- **Biomedical Image & Signal Processing:** Task-based / Resting-state Brain mapping, Neuroimaging using MRI, DTI, fMRI
- **Computational Neuroscience**
- **Biomedical Pattern Recognition**
- **Machine Learning with Medical Approach**
- **Cognitive Neuroscience:** Studying human brain functional organization with network-based approaches in Cognition, Language, Learning & Memory

EDUCATION

University of California, Irvine Irvine, CA
Ph.D. Student in Biomedical Engineering Sep 2023 – Jun 2025

- Completed doctoral-level coursework and research rotations focusing on neuroimaging and machine learning models.

Master of Science (M.Sc.) in Biomedical Engineering Sep 2019 – Jan 2022

- **Specialization:** Neuroimaging, Brain Decoding & Machine Learning
- Focused on computational modeling of brain data, advanced signal analytics, and machine learning techniques applied to medical diagnostics.

Bachelor of Engineering (B.E.) in Biomedical Engineering Sep 2015 – Feb 2019

- **Core Foundations:** Coursework in biomedical instrumentation, signals and systems, biostatistics, and foundational programming.

TECHNICAL SKILLS

Programming Languages: Python, MATLAB, Bash, Command Line, Git

Neuroimaging & Analysis: fMRI, DTI, Network Analysis, Brain Connectivity, Time Series Analysis, Surface Analysis

Frameworks & Tools: PyTorch, NumPy, Pandas, Scikit-Learn, Streamlit, Jupyter, MRtrix, FSL, FreeSurfer, BIDS

Statistics & Machine Learning: Linear Models, Mixed Models, Prediction, Classification, Clustering

Languages: English, Persian

RESEARCH & WORK EXPERIENCE

Graduate Researcher Sep 2024 – Jun 2025
Sanger Lab (Prof. Terence David Sanger) – UC Irvine

- **Neuroimaging & Tractography:** Processed structural MRI/DTI datasets, utilizing FreeSurfer to parcellate cortical and subcortical regions and mapping contralateral fiber connections between targeted brain areas using MRtrix.

Graduate Research Rotational Student Jan 2024 – Jul 2024
Visual Perception & Neuroimaging Lab (VPNL) (Prof. Emily Grossman) – UC Irvine

- Completed a 6-month lab rotation (Winter and Spring quarters) in the Visual Perception & Neuroimaging Lab (VPNL), focusing on human subject fMRI methodologies, signal preprocessing, and pattern classification.

Graduate Research Student Sep 2023 – Sep 2024
Biomedical Engineering Department – UC Irvine

- Conducted interdisciplinary research rotations across departmental units.

Teaching Assistant Oct 2024 – Jun 2025
Samueli School of Engineering – UC Irvine

- Evaluated coursework and provided constructive feedback to support academic development in engineering subjects.

Research Assistant

Aug 2022 – May 2023

Multiscale Bioimaging & Bioinformatics (MBB) Lab – Tulane University

- Conducted computational research supervised by Prof. Yu-Ping Wang.
- Designed machine learning and deep learning models to classify complex fMRI time-series data.

Teaching Assistant

Feb 2020 – Jul 2020

Graduate Level Courses: Biological Systems Modeling & Biological Signal Processing

- Provided weekly assignments and summaries of core course concepts.
- Received student assignments and projects, graded coursework, and provided constructive feedback.

RESEARCH PROJECTS

Structural Connectivity Analysis

DTI & Graph Theory

- Quantified structural brain networks using Diffusion Tensor Imaging pipelines to analyze topological organization and network connectivity metrics.

Unsupervised Pattern Discovery

Machine Learning & fMRI

- Applied clustering algorithms and unsupervised latent variable learning to uncover hidden functional brain states across multi-subject fMRI datasets.

fMRI Signal Modeling & Brain Decoding

FIR Modeling & Classification

- Reconstructed temporal fMRI dynamics using Finite Impulse Response (FIR) modeling and classified cognitive states during cued and uncued tasks.

Evaluation of Brain Connectivity in Parkinson's Disease

Master's Thesis Research

- Investigated resting-state functional connectivity via Independent Component Analysis (ICA) and Granger Causality Analysis (GCA) mapped to the AAL atlas.

PUBLICATIONS & PREPRINTS

- **Ahmadimehr, S.** et al. "Investigation of linear & non-linear connectivity." *Research Preprint*. [Link to Paper]
- **Ahmadimehr, S.** et al. "Alteration of resting-state brain networks in Parkinson's disease: A multi-aspect study of functional and effective connectivity." *Research Preprint*. [Link to Paper]

REFERENCES

Prof. Terence David Sanger

Electrical Engineering and Computer Science
Vice President, Chief Scientific Officer, CHOC
Vice Chair of Research for Pediatrics, School of Medicine
Staff Physician, CHOC Department of Neurology
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Prof. Emily Grossman

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Prof. Yu-Ping Wang

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Prof. Nader Jafarnia Dabanloo

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